

The Strömgren Sphere: Formation, Evolution and its Role in the Very Early Universe

A concept-and-equation summary built from the papers in `references/`

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Abstract

This note collects the physical concepts and the equations that govern the formation and the evolution of a Strömgren sphere, from the classical stellar H II region (Strömgren, 1939) to the cosmological H II regions (“proximity zones”, “ionized bubbles”) that surround the first stars, the first quasars and the JWST-era broad-line sources at $z \gtrsim 6$. Every physical statement below is tied to one of the references stored in `references/` and listed in `references.bib`. Section 7 reports the internal-consistency checks (analytic, with SYMPY, and numerical) that were run on the equations quoted here; the scripts are `check_consistency.py` and `alpha_HI_recombination_fits.py`.

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1 The classical Strömgren sphere

1.1 The idea

Strömgren (1939) showed that the Balmer emission observed by Struve & Elvey in the diffuse interstellar medium can be understood if the ionizing radiation of an O-type star is absorbed inside a well-defined volume: “*Balmer-line emission should be limited to certain rather sharply bounded regions in space surrounding O-type stars or clusters of O-type stars*”, with diameters of order 200 pc for the interstellar densities he assumed (Strömgren, 1939). The sharpness of the boundary is a consequence of the very large photoionization cross-section at the Lyman limit: the transition layer is optically thin in size compared to the ionized volume.

1.2 Ionization balance and the Strömgren radius

In a static, uniform, pure-hydrogen medium, a steady source emitting $\dot{N}_{\text{ion}}$ hydrogen-ionizing photons per second builds an ionized sphere whose radius follows from the requirement that *every ionizing photon is balanced by one recombination* inside the sphere (Cen and Haiman, 2000, eq. 1):

$$R_S = \left[\frac{3 \dot{N}_{\text{ion}}}{4\pi \alpha_B \langle n_H^2 \rangle} \right]^{1/3}, \quad (1)$$

where α_B is the hydrogen recombination coefficient at $T = 10^4$ K and $\langle n_H^2 \rangle$ is the mean squared hydrogen density inside R_S (Cen and Haiman, 2000). Equation (1) is the defining relation of the Strömgren sphere.

1.3 Case A, Case B and the on-the-spot approximation

The choice of recombination coefficient is not cosmetic. Recombinations directly to the ground state ($n = 1$) produce a fresh Lyman-continuum photon. The *on-the-spot* (OTS) approximation assumes that such a photon is re-absorbed close to where it was emitted, so that ground-state recombinations can simply be omitted; one then uses Case B, α_B , the sum over recombinations to all levels *except* the ground state (Nebrin, 2023). If instead the diffuse photons escape, the correct coefficient is Case A, α_A , the sum over all levels (Nebrin, 2023). Standard values are

$$\alpha_B(10^4 \text{ K}) = 2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1} \quad (\text{Shapiro et al., 2006}), \quad (2)$$

$$\alpha_A(T) \simeq 4.2 \times 10^{-13} T_4^{-0.7} \text{ cm}^3 \text{ s}^{-1} \quad (\text{Gnedin and Madau, 2022; Choudhury, 2022}), \quad (3)$$

with $T_4 \equiv T/10^4$ K. Nebrin (2023) derived an effective coefficient that interpolates between the two limits for a cloud of Lyman-limit optical thickness τ_{cl} ,

$$\langle \alpha_{\text{eff}} \rangle = \alpha_A - (\alpha_A - \alpha_B) f(\tau_{\text{cl}}), \quad (4)$$

with $f(\tau_{\text{cl}}) \ll 1$ (hence $\langle \alpha_{\text{eff}} \rangle \simeq \alpha_A$) for optically thin clouds and $f(\tau_{\text{cl}}) \simeq 1$ (hence $\langle \alpha_{\text{eff}} \rangle \simeq \alpha_B$) for optically thick clouds (Nebrin, 2023). Adopting Case A where Case B applies (or vice versa) without solving for the diffuse field introduces systematic errors in the predicted ionization state (Nebrin, 2023).

1.4 Temperature dependence of the H I recombination coefficient

Equation (1) is usually evaluated at the fiducial $T = 10^4$ K, but α_B is a steep function of temperature, and Sec. 4 shows that the gas behind a cosmological I-front can be anywhere between $\sim 1.7 \times 10^4$ and 3×10^4 K. The relevant expressions are collected here; all of them are fits to detailed collisional-radiative calculations, not independent theory. They are implemented in `alpha_HI_recombination_fits.py`.

Case B (recommended closed form). Hui and Gnedin (1997) give, in their Appendix A, a fit to the data of Ferland et al. (1992) that is quoted as accurate to 0.7% over $1 \text{ K} \leq T \leq 10^9 \text{ K}$:

$$\alpha_B(T) = 2.753 \times 10^{-14} \frac{\lambda^{1.500}}{[1 + (\lambda/2.740)^{0.407}]^{2.242}} \text{ cm}^3 \text{ s}^{-1} \quad \lambda \equiv \frac{2 T_{\text{HI}}}{T}, \quad (5)$$

with $T_{\text{HI}} = 157807 \text{ K}$ the H I ionization threshold expressed in temperature units (Hui and Gnedin, 1997). This is the single most convenient expression for Strömgren-sphere work: it is analytic, has the correct asymptotic behaviour, and is smooth across the whole nebular and IGM range.

Case A, same source.

$$\alpha_A(T) = 1.269 \times 10^{-13} \frac{\lambda^{1.503}}{[1 + (\lambda/0.522)^{0.470}]^{1.923}} \text{ cm}^3 \text{ s}^{-1}, \quad (6)$$

quoted as accurate to 2% over $3 \text{ K} \leq T \leq 10^9 \text{ K}$ (Hui and Gnedin, 1997).

Total (Case A) rate from atomic-data fits. Verner and Ferland (1996) fitted the total radiative recombination rate of all H-like ions over $3 \text{ K} - 10^{10} \text{ K}$ with

$$\alpha_r(T) = a \left[\sqrt{T/T_0} \left(1 + \sqrt{T/T_0} \right)^{1-b} \left(1 + \sqrt{T/T_1} \right)^{1+b} \right]^{-1}, \quad (7)$$

which by construction gives $\alpha_r \propto T^{-1/2}$ for $T \ll T_0 \ll T_1$ and $\alpha_r \propto T^{-3/2}$ for $T \gg T_1 \gg T_0$ (Verner and Ferland, 1996). Their Table 1 lists, for H I,

$$a = 7.982 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}, \quad b = 0.7480, \quad T_0 = 3.148 \text{ K}, \quad T_1 = 7.036 \times 10^5 \text{ K}, \quad (8)$$

with rms error $< 0.5\%$ (maximum $< 1\%$) for hydrogenic species (Verner and Ferland, 1996). Hydrogenic rates for any charge follow from $\alpha_r(Z, T) = Z \alpha_r(1, T/Z^2)$ (Verner and Ferland, 1996). The same functional form, with $b \rightarrow b + b_1 \exp(T_2/T)$ for some ions, is what Mao and Kaastra (2016) adopt for the level-resolved rates, so eq. (7) is the current standard shape.

Older analytic forms, for provenance. Two expressions still widely encountered are, first, the hydrogenic formula of Seaton (1959) as quoted in eq. (6) of Verner and Ferland (1996),

$$\alpha_r(Z, T) = 5.197 \times 10^{-14} Z \lambda_S^{1/2} \left[0.4288 + \frac{1}{2} \ln \lambda_S + 0.469 \lambda_S^{-1/3} \right] \text{ cm}^3 \text{ s}^{-1}, \quad \lambda_S = \frac{157890 Z^2}{T(\text{K})}, \quad (9)$$

which Verner and Ferland (1996) note is *not* valid for $T > 10^6 Z^2 \text{ K}$ because its high-temperature asymptote is wrong; and second, the four-parameter form of Péquignot, Petitjean & Boisson (1991), also quoted by Verner and Ferland (1996),

$$\alpha_r(T) = 10^{-13} z \frac{a t^b}{1 + c t^d} \text{ cm}^3 \text{ s}^{-1}, \quad t = 10^{-4} T / z^2, \quad (10)$$

valid only over $40 z^2 \leq T \leq 4 \times 10^4 z^2 \text{ K}$ (Verner and Ferland, 1996).

Polynomial form used in primordial-chemistry networks. Abel et al. (1997) fit the same Ferland et al. (1992) data with a ninth-order polynomial in $\ln T$,

$$k_2 = \exp \left[\sum_{i=0}^9 c_i (\ln T)^i \right] \text{ cm}^3 \text{ s}^{-1}, \quad (11)$$

with $c_0 = -28.6130338$, $c_1 = -0.72411256$, $c_2 = -2.02604473 \times 10^{-2}$, $c_3 = -2.38086188 \times 10^{-3}$, $c_4 = -3.21260521 \times 10^{-4}$, $c_5 = -1.42150291 \times 10^{-5}$, $c_6 = +4.98910892 \times 10^{-6}$, $c_7 = +5.75561414 \times 10^{-7}$, $c_8 = -1.85676704 \times 10^{-8}$, $c_9 = -3.07113524 \times 10^{-9}$ (Abel et al., 1997). **Unit trap:** the caption of their Table 1 states that “the temperatures are in eV unless stated otherwise”, so the argument of eq. (11) is $\ln(T/\text{eV})$, not $\ln(T/\text{K})$. Using kelvin gives a value ~ 4 orders of magnitude too small; with T in eV, eq. (11) reproduces eq. (6) to within 3% over 10^2 – 10^6 K, i.e. it is a *Case A* rate (see check 10 in Sec. 7).

Power-law approximations and their validity. Differentiating eq. (5) numerically gives a local logarithmic slope $d \ln \alpha_B / d \ln T = -0.787$, -0.834 and -0.888 at $T = 5 \times 10^3$, 10^4 and 2×10^4 K, and for Case A $d \ln \alpha_A / d \ln T = -0.685$, -0.714 and -0.751 at the same temperatures. The power laws

$$\alpha_B(T) \simeq 2.59 \times 10^{-13} T_4^{-0.83} \text{ cm}^3 \text{ s}^{-1}, \quad (12)$$

$$\alpha_A(T) \simeq 4.2 \times 10^{-13} T_4^{-0.70} \text{ cm}^3 \text{ s}^{-1}, \quad (13)$$

are accurate to 2.1% and 2.5% respectively over 5×10^3 – 2×10^4 K, degrading to 5.2% and 2.5% over 5×10^3 – 3×10^4 K and to 25% and 19% if stretched over 10^3 – 10^5 K. Equation (13) is exactly the expression quoted by Gnedin and Madau (2022) and Choudhury (2022); there is *no* single power law that is simultaneously accurate for Case A and Case B, because the two have different slopes. Since $R_S \propto \alpha_B^{-1/3}$ (eq. 1), an error δ in α_B propagates to only $\delta/3$ in R_S , so eqs. (12)–(13) are adequate for radii; they are not adequate for photon-budget or emission-measure work, where α_B enters linearly.

Regime of validity, and what these fits do *not* include. All the expressions above are for pure radiative recombination in the low-density limit, with a Maxwell–Boltzmann electron energy distribution. Two caveats matter for early-Universe work. (i) Case B is itself density-dependent through the collisional redistribution of the excited levels: Storey and Sochi (2014) tabulate the hydrogen total Case B coefficient as $\alpha(N_e, T_e, \kappa)$, i.e. explicitly as a function of electron density as well as temperature, so eq. (5) should be read as the low-density limit. (ii) If the electron energy distribution departs from Maxwellian — as it may where non-thermal particles inject energy — the recombination coefficients change: Storey and Sochi (2014) compute them for a κ -distribution, recovering the Maxwell–Boltzmann values only as $\kappa \rightarrow \infty$. Neither effect is captured by eqs. (5)–(13).

1.5 Time-dependent growth: the recombination clock

Before equilibrium is reached, the ionization front (I-front) sweeps up neutral gas. Writing the photon budget as a balance between injection and recombination inside the ionized volume gives, for a static uniform medium of clumping factor C ,

$$\frac{dR^3}{dt} = \frac{3 \dot{N}_{\text{ion}}}{4\pi n_{\text{H}}} - \alpha_B C n_{\text{H}} R^3, \quad (14)$$

whose solution is the familiar exponential approach to the Strömberg radius,

$$R(t) = R_S (1 - e^{-t/t_{\text{rec}}})^{1/3}, \quad t_{\text{rec}} = (C \alpha_B n_{\text{H}} \chi_{\text{eff}})^{-1}, \quad (15)$$

in agreement with the non-dimensional variable $w \equiv t/t_{\text{rec}} = tC\alpha_B n_H \chi_{\text{eff}}$ and the solution $y = 1 - e^{-w}$ of Shapiro et al. (2006); χ_{eff} accounts for the extra free electrons contributed by ionized helium. The recombination time is therefore the clock of the problem: the sphere is essentially complete after a few t_{rec} .

1.6 Relativistic (finite light-speed) corrections

Shapiro et al. (2006) showed that the non-relativistic equation (15) can be badly wrong at early times, because the I-front is initially so fast that a naive treatment yields apparent superluminal speeds. The controlling dimensionless parameter is

$$q \equiv \frac{R_S}{c t_{\text{rec}}}, \quad (16)$$

the light-crossing time of the Strömgren radius in units of the recombination time (Shapiro et al., 2006). For $q > 1$ the expansion is relativistic all the way out to R_S and the time to reach the Strömgren radius is delayed by a factor $\sim q$ relative to the non-relativistic answer (Shapiro et al., 2006). This is the regime relevant to bright cosmological sources, where t_{rec} is enormous (Sec. 3).

2 Ionization fronts: R-type, D-type, and the dynamical phase

2.1 Front classification

Around a newly switched-on source the I-front first moves supersonically with respect to both the neutral and the ionized gas: it is a weak *R-type* front, and the gas is ionized essentially without being moved (Shapiro et al., 2006). As the front decelerates (its speed drops roughly as the volume grows) it approaches the sound speed of the ionized gas, c_i , and transforms into a subsonic *D-type* front preceded by a shock; from then on the hot, over-pressured ionized bubble expands hydrodynamically into the surrounding medium (Shapiro et al., 2006). The R-type phase is short-lived (at most a few t_{rec}), whereas the D-type phase is long-lived, and during it $v_I \ll c$ so that relativistic corrections are negligible (Shapiro et al., 2006).

2.2 D-type expansion: Spitzer and Hosokawa–Inutsuka solutions

In the thin-shell approximation, equating the pressure of the shocked neutral shell to the ram pressure of the undisturbed gas gives the “Raga-I” equation, whose early-time limit is the classical *Spitzer solution* (Bisbas et al., 2015, eq. 9):

$$R_{\text{Sp}}(t) = R_{\text{St}} \left(1 + \frac{7}{4} \frac{c_i t}{R_{\text{St}}} \right)^{4/7}, \quad (17)$$

with R_{St} the initial Strömgren radius of eq. (1). Including the inertia of the shocked shell (Hosokawa & Inutsuka, “Raga-II”) yields

$$\dot{R}_{\text{HI}}(t) = c_i \sqrt{\frac{4}{3} \left(\frac{R_{\text{St}}}{R_{\text{HI}}} \right)^{3/2} - \frac{\mu_i T_o}{\mu_o T_i}}, \quad (18)$$

(Bisbas et al., 2015, eq. 12). Expansion stops when the ionized gas reaches pressure equilibrium with the ambient medium, at the *stagnation radius*

$$R_{\text{STAG}} = \left(\frac{c_i}{c_o} \right)^{4/3} R_{\text{St}}, \quad (19)$$

where c_o is the sound speed of the neutral gas (Bisbas et al., 2015, eqs. 14 and 22). Because $c_i/c_o \sim 4\text{--}10$, the final H II region is one to two orders of magnitude larger in volume than the

initial Strömgren sphere. [Bisbas et al. \(2015\)](#) showed with the STARBENCH code comparison that a semi-empirical combination of the early- and late-time solutions reproduces high-resolution radiation-hydrodynamics simulations to $\lesssim 2\%$, and is a better code-validation benchmark than the Spitzer solution alone.

2.3 The first H II regions: Population III stars in minihaloes

In the primordial case the ambient density profile is steeply declining, and the ordering of the two front types is inverted. [Kitayama et al. \(2004\)](#) found that the H II region of a Pop III star begins as a slowly expanding weak D-type front near the centre and then, once the front outruns the declining density profile, converts to a rapidly propagating R-type front through the outer envelope; this D→R transition is the critical condition for complete ionization of a cosmologically relevant halo. For halo masses $\lesssim 10^6 M_\odot$ the transition occurs within a few 10^5 yr, the escape fractions of both ionizing and photodissociating photons exceed 80%, and a supersonic shock evacuates the halo, lowering the mean density to $\lesssim 1 \text{ cm}^{-3}$ in a few Myr; for $\gtrsim 10^7 M_\odot$ haloes the front stays D-type for the whole stellar lifetime, the H II region remains confined inside the virial radius and the escape fraction is essentially zero ([Kitayama et al., 2004](#)). [Alvarez et al. \(2006\)](#), with 3D ray tracing, described the same physics as a “champagne flow” in which the early D-type front detaches from the shock and runs ahead as an R-type front; they obtained $0.7 \lesssim f_{\text{esc}} \lesssim 0.9$ for $80 \lesssim M_\star/M_\odot \lesssim 500$ and a ratio of ionized gas mass to stellar mass of $\sim 6 \times 10^4$, i.e. roughly half the ionizing photons released per stellar baryon are used to ionize the intergalactic gas. They also found that neighbouring minihaloes *trap* the I-front, so that their cores remain neutral ([Alvarez et al., 2006](#)).

3 Cosmological Strömgren spheres

3.1 Equation of motion in an expanding universe

For a cosmological source the Strömgren problem acquires two new ingredients: the Hubble expansion dilutes the gas, and the intergalactic medium (IGM) is clumpy. Following [Shapiro and Giroux \(1987\)](#), the proper radius R_i of the I-front obeys ([Cen and Haiman, 2000](#), eq. 3)

$$\frac{dR_i^3}{dt} = 3H(z) R_i^3 + \frac{3\dot{N}_{\text{ph}}}{4\pi \langle n_{\text{H}} \rangle} - C_{\text{HII}} \langle n_{\text{H}} \rangle \alpha_{\text{B}} R_i^3, \quad (20)$$

where the three terms on the right-hand side are, in order, the Hubble expansion, the ionizations by newly produced photons, and the recombinations, and

$$C_{\text{HII}} \equiv \frac{\langle n_{\text{H}}^2 \rangle}{\langle n_{\text{H}} \rangle^2} \quad (21)$$

is the mean clumping factor of the ionized gas inside R_i ([Cen and Haiman, 2000](#)). A more careful definition weights by the electron density, $C \equiv \langle n_{\text{HII}} n_e \rangle / (\bar{n}_{\text{HII}} \bar{n}_e)$, restricted to the ionized regions ([Choudhury, 2022](#), eq. 64); $C = 1$ for a uniform medium. The clumping factor is the standard way of encoding IGM inhomogeneity as an enhanced effective recombination rate ([Miralda-Escude et al., 2000](#); [Choudhury, 2022](#)), and it is not straightforward to model self-consistently because it depends on the density structure and on how I-fronts propagate into dense regions ([Choudhury, 2022](#)).

3.2 The photon-counting limit: why high- z spheres never reach equilibrium

The key qualitative difference with respect to the stellar case is that at $z \lesssim 10$ the *recombination time of the IGM is comparable to or longer than the Hubble time, and enormously longer than the quasar lifetime*, so the equilibrium radius R_{S} is large but is never actually reached before the

source turns off (Cen and Haiman, 2000). (For $C = 1$ we find $t_{\text{rec}}/H^{-1}(z) \simeq 1.2, 0.8$ and 0.6 at $z = 6, 8$ and 10 ; see check 7 in Sec. 7. The statement is therefore safe with respect to t_Q , which is what matters here, and marginal with respect to the Hubble time at the highest redshifts.) Dropping the recombination term in eq. (20) leaves a pure photon-counting result for a source of age t_Q (Cen and Haiman, 2000, eq. 2):

$$R_{t_Q} = \left[\frac{3 \dot{N}_{\text{ph}} t_Q}{4\pi \langle n_{\text{H}} \rangle} \right]^{1/3}. \quad (22)$$

Including the neutral fraction x_{HI} of the gas that must be ionized, the fiducial scaling for a luminous $z \sim 6$ quasar is (Mesinger and Haiman, 2004)

$$R_{\text{S}} \simeq 7.7 x_{\text{HI}}^{-1/3} \left(\frac{\dot{N}_Q}{6.5 \times 10^{57} \text{ s}^{-1}} \right)^{1/3} \left(\frac{t_Q}{2 \times 10^7 \text{ yr}} \right)^{1/3} \left(\frac{1+z_Q}{7.28} \right)^{-1} \text{ Mpc (proper)}. \quad (23)$$

The three exponents encode the whole physics: $R \propto \dot{N}^{1/3}$ and $R \propto t_Q^{1/3}$ from photon counting, and $R \propto (1+z)^{-1}$ from $\langle n_{\text{H}} \rangle \propto (1+z)^3$. Unlike the stellar case, the cosmological sphere keeps evolving even at constant luminosity, because of the Hubble expansion, of the evolving density and clumping around the source, and of the continued accumulation of ionizing photons (Cen and Haiman, 2000).

3.3 From individual spheres to reionization

Globally, the same photon budget written for the volume filling factor Q_{HII} of ionized gas gives the standard reionization equation (Choudhury, 2022, eq. 66)

$$\frac{dQ_{\text{HII}}}{dt} = \frac{\dot{n}_{\gamma}}{\bar{n}_{\text{H}}} - Q_{\text{HII}} a^{-3} C \alpha_{\text{R}}(T) \bar{n}_{\text{e}}, \quad (24)$$

i.e. a source term set by the ionizing emissivity and a sink term set by the clumping-enhanced recombination rate; $\chi_{\text{He}} \approx 1.08$ (1.16) for singly (doubly) ionized helium with $Y = 0.24$ (Choudhury, 2022). Reionization proceeds inside-out: isolated ionized bubbles form around the first sources, then grow, overlap and finally fill the IGM, leaving neutral gas only in dense pockets (Chakraborty and Choudhury, 2025; Cen and Haiman, 2000).

The size *distribution* of these bubbles is not set by the Strömgren radius of individual sources but by the clustering of the sources. Furlanetto et al. (2004) associated each ionized region with a large-scale density fluctuation that can “self-ionize”, $m_{\text{ion}} = \zeta m_{\text{gal}}$, i.e.

$$f_{\text{coll}}(\delta_m, m, z) \geq \zeta^{-1}, \quad (25)$$

and used the excursion-set formalism with the resulting density barrier $\delta_x(m, z) = \delta_c(z) - \sqrt{2} K(\zeta) [\sigma_{\text{min}}^2 - \sigma^2(m)]^{1/2}$, with $K(\zeta) = \text{erf}^{-1}(1 - \zeta^{-1})$, to obtain the bubble size distribution (Furlanetto et al., 2004). In this model bubbles are already several comoving Mpc across by the time the ionized fraction reaches $\bar{Q} = 0.5$ (Furlanetto et al., 2004). Modern radiation-hydrodynamic simulations confirm that bubble sizes correlate with the local overdensity, and that the local reionization redshift z_{reion} is a good proxy for bubble sizes only up to $R_{\text{eff}} \lesssim 1 \text{ cMpc}$ (Neyer et al., 2023).

4 Thermal structure: the temperature left behind by the I-front

The Strömgren sphere is not only an ionization structure but also a heating structure: the excess energy of each ionizing photon above 13.6 eV is deposited as heat when the front passes. The post-front temperature T_{reion} is a key, and uncertain, ingredient of reionization models (D’Aloisio et al., 2018). Its main determinant is *the I-front speed*, not the spectral index of the source (D’Aloisio et al., 2018; Zhu and Gnedin, 2025):

- slow fronts ($v_I \sim 50\text{--}2000 \text{ km s}^{-1}$) give $T \sim 17,000\text{--}22,000 \text{ K}$, typical of the first half of reionization (D’Aloisio et al., 2018; Zhu and Gnedin, 2025);
- fast fronts near the overlap phase ($v_I \sim 10^4 \text{ km s}^{-1}$) reach $T_{\text{reion}} \sim 25,000\text{--}30,000 \text{ K}$ (D’Aloisio et al., 2018; Zhu and Gnedin, 2025);
- the median post-ionization temperature plateaus at $T \sim 2 \times 10^4 \text{ K}$ for $v_I > 3000 \text{ km s}^{-1}$ (Zhu and Gnedin, 2025).

The usual assumption that all baryonic species (e^- , H I, H II, He I, He II) instantaneously share the deposited energy breaks down for fast fronts: Zeng and Hirata (2020) solved the non-equilibrium energy-transfer problem and found that equilibration is a good approximation for $v_I \lesssim 10^9 \text{ cm s}^{-1}$, while T_{re} is lower than the equilibrium value by up to 19.7% at $3 \times 10^9 \text{ cm s}^{-1}$ and 30.8% at $10^{10} \text{ cm s}^{-1}$. Since electrons carry the photoelectron energy first and then share it with the ions, the electron–ion energy transfer rate is what sets this deviation (Zeng and Hirata, 2020). Uncertainties in T_{reion} of 15,000 vs. 25,000 K propagate into $\sim 50\%$ differences in the IGM temperature at $z \simeq 5.5$ for gas reionized at $z_{\text{re}} = 9$ (D’Aloisio et al., 2018), which is directly relevant to $z > 5$ Ly α forest thermal measurements.

5 Observational manifestations at high redshift

5.1 The Strömgren surface in quasar spectra

A bright quasar embedded in a largely neutral IGM at $z \gtrsim 6$ is surrounded by a large cosmological Strömgren sphere; its spectrum then shows a sharp increase in resonant Lyman-line absorption at wavelengths approaching the boundary of the sphere along the line of sight (Mesinger and Haiman, 2004). Using *simultaneously* the Ly α and Ly β regions to gain dynamic range, Mesinger and Haiman (2004) detected the boundary of the Strömgren sphere of SDSS J1030+0524 ($z = 6.28$) at a proper distance of $6.0 \pm 0.2 \text{ Mpc}$, inferred $x_{\text{HI}} \gtrsim 0.2$ beyond it, and measured an ionizing output of $(5.2 \pm 2.5) \times 10^{56} \text{ photons s}^{-1}$. The underlying spectra of the two first Gunn–Peterson-trough quasars, and the fact that the Strömgren sphere of SDSS J1148+5251 is significantly smaller, were presented by White et al. (2003).

The Lyman- α line profile itself carries information: for a source inside a neutral IGM the damping wing of the intervening H I suppresses the line, whereas a sufficiently large H II region lets Ly α photons redshift out of resonance before reaching the boundary (Cen and Haiman, 2000). The red side of the line then measures the quasar lifetime and the blue side the IGM density fluctuations at the quasar redshift (Cen and Haiman, 2000).

5.2 Proximity zones and quasar lifetimes

Operationally, the observed quantity is the *proximity zone* size R_p , defined as the distance at which the smoothed, continuum-normalized quasar spectrum first drops below the 10% transmitted flux level (Eilers et al., 2017). This definition probes locations where the quasar’s ionizing radiation still exceeds the expected UV background by about an order of magnitude, which makes R_p comparatively *insensitive* to the ambient IGM neutral fraction (Eilers et al., 2017). This is the main caveat when using near zones as a reionization probe: Bolton and Haehnelt (2007) showed with synthetic spectra that the scatter induced by density fluctuations and by the filtering of ionizing radiation is so large that the data then available could not distinguish a significantly neutral from a highly ionized IGM just above $z = 6$, and that samples of several tens of high-resolution Ly α + Ly β near zones are needed to decide whether $\langle f_{\text{HI}} \rangle_V$ is above or below 10%.

What R_p *does* measure well is the quasar age, because the IGM has a finite response time to the radiation (Eilers et al., 2020). The observational picture is:

- R_p evolves only shallowly with redshift, $\propto (1+z)^{-1.44}$ over $5.77 \leq z \leq 6.54$, in agreement with radiative-transfer simulations irrespective of the IGM ionization state (Eilers et al., 2017);
- a few quasars have exceptionally small zones ($R_p < 1$ pMpc), most compellingly explained by UV-luminous lifetimes $t_Q \lesssim 10^5$ yr (Eilers et al., 2017);
- a dedicated multi-wavelength survey found $5\% \lesssim f_{\text{young}} \lesssim 10\%$ of the high- z quasar population to be young, with $t_Q \lesssim 10^5$ yr (Eilers et al., 2020);
- the enlarged XQR-30 sample gives $R_p = 2\text{--}7$ pMpc for $M_{1450} \sim -27$ quasars at $5.8 < z < 6.6$, with the scatter indicating a large diversity of quasar lifetimes (Satyavolu et al., 2023);
- the largest homogeneous compilation to date, 59 quasars at $5.77 \leq z \leq 7.54$, gives $R_p \propto 10^{-0.4M_{1450}/2.87}$ (i.e. $R_p \propto L^{1/2.87}$) and $R_p \propto (1+z)^{-2.44}$, and identifies 13 small-zone quasars for which $t_Q \lesssim 10^4$ yr is the favoured explanation (Onorato et al., 2025).

Note that the measured luminosity scaling $L^{1/2.87} = L^{0.348}$ is remarkably close to the naive photon-counting exponent $1/3$ of eq. (22), whereas the redshift scaling is steeper than the $(1+z)^{-1}$ of eq. (23), which is expected since the latter holds at fixed x_{HI} and fixed t_Q (Onorato et al., 2025; Mesinger and Haiman, 2004).

5.3 Ionized bubbles around JWST galaxies and AGN

For galaxies, the same physics is inverted: one models the Ly α damping-wing absorption assuming a galaxy sitting inside an ionized bubble of radius R_b embedded in an IGM of neutral fraction x_{HI} . From 27 bright ($M_{\text{UV}} < -18.5$) JWST/NIRSpec galaxies at $7 < z < 12$, Umeda et al. (2023) obtained x_{HI} rising from $0.53^{+0.18}_{-0.47}$ at $z = 7.12$ to $0.92^{+0.08}_{-0.10}$ at $z = 9.91$, with $\log(R_b/\text{cMpc})$ falling from $1.67^{+0.14}_{-0.16}$ to $-0.69^{+1.49}_{-0.24}$ over the same interval; the inferred R_b can be up to a few dex larger than the cosmic-average analytic expectation at a given x_{HI} , but is compatible with the merged bubbles predicted around bright galaxies in simulations (Umeda et al., 2023). A Ly α line-shape fit that explicitly assumes a Strömgren sphere gives $R_b = 1.5^{+0.3}_{-0.2}$ pMpc around a nitrogen-rich AGN at $z = 8.63$ — a large bubble given that the Universe is nearly fully neutral at that redshift (Morishita et al., 2025). As quoted by Morishita et al. (2025), the predicted IGM damping-wing transmission at Ly α line centre is $< 10\%$ if the source sits in an ionized region smaller than 0.5 pMpc, versus $> 30\%$ for a region larger than 1 pMpc; this is why the detection of Ly α near the systemic velocity at $z > 8$ is itself evidence for a substantial Strömgren sphere.

5.4 Who blows the bubbles? Relevance for the ionizing photon budget

Whether the ionizing photons come from faint star-forming galaxies or from accreting black holes determines both the bubble morphology and the reionization history (Chakraborty and Choudhury, 2025; Gnedin and Madau, 2022). Mascia et al. (2026) carried out a systematic census of compact broad-line emitters (BLEs) at $4 \leq z \leq 7$ from JWST/NIRSpec: the bluest, compact BLEs show hot ionizing continua, high-ionization UV lines, elevated Ly α -emitter fractions ($X_{\text{Ly}\alpha} = 50\text{--}67\%$), a high ionizing photon production efficiency $\log(\xi_{\text{ion}}/\text{Hz erg}^{-1}) \sim 25.4$ and a weighted escape fraction $f_{\text{esc}} \sim 0.2$, whereas Little Red Dots have $f_{\text{esc}} \sim 0.01$. Their conclusion is that broad-line sources do *not* dominate reionization globally, but that, given their luminosity, their contribution can dominate *up to few-Mpc scales* (Mascia et al., 2026) — i.e. exactly on the scale of an individual Strömgren sphere.

Quantity	Expression	Reference
Strömgren radius (static)	$R_S = [3\dot{N}_{\text{ion}}/(4\pi\alpha_B\langle n_H^2 \rangle)]^{1/3}$	Cen and Haiman (2000)
Case B coefficient	$\alpha_B(10^4\text{K}) = 2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$	Shapiro et al. (2006)
Case A coefficient	$\alpha_A \simeq 4.2 \times 10^{-13} T_4^{-0.7} \text{ cm}^3 \text{ s}^{-1}$	Gnedin and Madau (2022)
Case B, full T dependence	$\alpha_B = 2.753 \times 10^{-14} \lambda^{1.500} [1 + (\lambda/2.740)^{0.407}]^{-2.242}, \lambda = 2T_{\text{HI}}/T$	Hui and Gnedin (1997)
Case A, full T dependence	$\alpha_A = 1.269 \times 10^{-13} \lambda^{1.503} [1 + (\lambda/0.522)^{0.470}]^{-1.923}$	Hui and Gnedin (1997)
Total rate, H-like ions	$\alpha_r = a[\sqrt{T/T_0}(1 + \sqrt{T/T_0})^{1-b}(1 + \sqrt{T/T_1})^{1+b}]^{-1}$	Verner and Ferland (1996)
Power-law approximations	$\alpha_B \simeq 2.59 \times 10^{-13} T_4^{-0.83}, \alpha_A \simeq 4.2 \times 10^{-13} T_4^{-0.70}$	Sec. 1.4
Effective coefficient	$\langle \alpha_{\text{eff}} \rangle = \alpha_A - (\alpha_A - \alpha_B)f(\tau_{\text{cl}})$	Nebrin (2023)
Growth to equilibrium	$R(t) = R_S(1 - e^{-t/t_{\text{rec}}})^{1/3}$	Shapiro et al. (2006)
Recombination time	$t_{\text{rec}} = (C\alpha_B n_H \chi_{\text{eff}})^{-1}$	Shapiro et al. (2006)
Relativistic parameter	$q = R_S/(c t_{\text{rec}})$	Shapiro et al. (2006)
D-type (Spitzer)	$R_{\text{Sp}} = R_{\text{St}}(1 + \frac{7}{4}c_i t/R_{\text{St}})^{4/7}$	Bisbas et al. (2015)
Stagnation radius	$R_{\text{STAG}} = (c_i/c_o)^{4/3} R_{\text{St}}$	Bisbas et al. (2015)
Cosmological I-front	$(\dot{R}_i^3) = 3H R_i^3 + \frac{3\dot{N}_{\text{ph}}}{4\pi\langle n_H \rangle} - C_{\text{HII}}\langle n_H \rangle \alpha_B R_i^3$	Cen and Haiman (2000); Shapiro and Giroux (1987)
Photon-counting radius	$R_{t_Q} = [3\dot{N}_{\text{ph}} t_Q/(4\pi\langle n_H \rangle)]^{1/3}$	Cen and Haiman (2000)
Fiducial high- z scaling	$R_S \simeq 7.7 x_{\text{HI}}^{-1/3} \dot{N}_{57}^{1/3} t_{Q,7}^{1/3} (1+z)^{-1} \text{ pMpc}$	Mesinger and Haiman (2004)
Reionization equation	$dQ_{\text{HII}}/dt = \dot{n}_\gamma/\bar{n}_H - Q_{\text{HII}}a^{-3}C\alpha_R\bar{n}_e$	Choudhury (2022)
Self-ionization barrier	$f_{\text{coll}} \geq \zeta^{-1}$	Furlanetto et al. (2004)
Post-front temperature	$T_{\text{reion}} \simeq 1.7\text{--}3.0 \times 10^4 \text{ K}$, set by v_{I}	D'Aloisio et al. (2018); Zhu and Gnedin (2025)
Proximity zone definition	first drop below 10% transmitted flux	Eilers et al. (2017)

Table 1: Master list of the relations discussed in this note.

6 Summary of the key equations and fiducial numbers

7 Internal consistency checks

The following checks were carried out with SYMPY 1.12 and NUMPY 1.26 (`check_consistency.py`); all of them passed.

- Eq. (15) solves eq. (14).** Substituting $R(t) = R_S(1 - e^{-\alpha_B C n_H t})^{1/3}$ with $R_S = [3\dot{N}_{\text{ion}}/(4\pi\alpha_B C n_H^2)]^{1/3}$ into $dR^3/dt = 3\dot{N}_{\text{ion}}/(4\pi n_H) - \alpha_B C n_H R^3$ gives a residual that SYMPY simplifies to exactly 0. This also confirms the identification $t_{\text{rec}} = (\alpha_B C n_H)^{-1}$ used in eq. (15) and by Shapiro et al. (2006).
- Eq. (17) solves the early-time Raga-I equation.** $\dot{R}_{\text{Sp}}/c_i - (R_{\text{St}}/R_{\text{Sp}})^{3/4}$ simplifies to 0, i.e. the Spitzer solution is the exact solution of eq. (8) of Bisbas et al. (2015) once the small $\mu_i T_o/\mu_o T_i$ term is dropped, as stated there.
- Eq. (19) reproduces the numbers quoted by Bisbas et al. (2015).** With $T_i = 10^4 \text{ K}$, $\mu_i = 0.5$ and $T_o = 10^3 \text{ K}$, $\mu_o = 1$ one gets $c_i = 12.85 \text{ km s}^{-1}$ and $c_o = 2.87 \text{ km s}^{-1}$ (their

values), and $R_{\text{STAG}} = (c_i/c_o)^{4/3} \times 0.314 \text{ pc} = 2.31 \text{ pc}$, identical to their quoted stagnation radius.

4. **Eq. (23) is the photon-counting radius eq. (22).** Evaluating $[3\dot{N}_Q t_Q / (4\pi n_{\text{H}} x_{\text{HI}})]^{1/3}$ with $\dot{N}_Q = 6.5 \times 10^{57} \text{ s}^{-1}$, $t_Q = 2 \times 10^7 \text{ yr}$, $x_{\text{HI}} = 1$, $z = 6.28$ and the cosmology of Mesinger and Haiman (2004) ($\Omega_b = 0.044$, $h = 0.71$, $Y = 0.24$; $n_{\text{H}} = 7.31 \times 10^{-5} \text{ cm}^{-3}$) gives 7.70 proper Mpc, exactly the coefficient quoted in their eq. (23).
5. **Proper/comoving conversion of the measured Strömgren surface.** Their best-fit $R_S = 44 \text{ cMpc}$ at $z = 6.28$ corresponds to $44/(1+z) = 6.04 \text{ pMpc}$, consistent with the $6.0 \pm 0.2 \text{ pMpc}$ quoted in their abstract (Mesinger and Haiman, 2004).
6. **Order of magnitude of eq. (1) for a stellar source.** With $\dot{N}_{\text{ion}} = 10^{49} \text{ s}^{-1}$ and $n_{\text{H}} = 1 \text{ cm}^{-3}$, eq. (1) gives $R_S = 68 \text{ pc}$, i.e. a diameter of 136 pc, of the same order as the “about 200 parsecs” quoted by Strömgren (1939) (the exact value depends on the stellar temperature, radius and gas density he adopted). For $n_{\text{H}} = 100 \text{ cm}^{-3}$ the same source gives $R_S = 3.2 \text{ pc}$ and $t_{\text{rec}} = 1.2 \text{ kyr}$, so a compact H II region reaches equilibrium essentially instantaneously — consistent with the classical, static interpretation.
7. **Why the cosmological sphere is *not* in equilibrium.** With $\chi_{\text{He}} = 1.08$ and $C = 1$, $t_{\text{rec}} = 1.74, 0.82$ and 0.45 Gyr at $z = 6, 8$ and 10 respectively, i.e. comparable to the Hubble time $1/H(z) = 1.43, 0.98$ and 0.73 Gyr , and four to five orders of magnitude longer than the quasar lifetimes $t_Q \sim 10^4\text{--}10^7 \text{ yr}$ inferred from proximity zones (Eilers et al., 2017, 2020; Onorato et al., 2025). This is a quantitative confirmation of the statement of Cen and Haiman (2000) that the equilibrium Strömgren radius is never reached at high redshift, and hence that eq. (22) — not eq. (1) — is the relevant relation. Caveat: at $C \gtrsim 2$ and $z \gtrsim 8$, t_{rec} becomes shorter than $1/H(z)$, so the recombination term of eq. (20) cannot be neglected for long-lived sources in dense environments.
8. **Luminosity scaling.** The empirical $R_p \propto L^{1/2.87} = L^{0.348}$ of Onorato et al. (2025) is within 5% of the photon-counting exponent $1/3 = 0.333$ of eq. (22), so the two are mutually consistent.
9. **I-front speed regimes.** The plateau velocity $3000 \text{ km s}^{-1} = 3 \times 10^8 \text{ cm s}^{-1}$ of Zhu and Gnedin (2025) lies *below* the 10^9 cm s^{-1} threshold above which Zeng and Hirata (2020) find departures from thermal equilibration, so the two results are consistent: the $\sim 2 \times 10^4 \text{ K}$ plateau is in the regime where the equilibrium assumption holds, and the 19.7–30.8% reductions of Zeng and Hirata (2020) apply only to the fastest fronts.
10. **Mutual consistency of the H I recombination fits of Sec. 1.4.** Evaluated at $T = 10^4 \text{ K}$, eq. (5) gives $\alpha_B = 2.5918 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$, reproducing the $2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ quoted by Shapiro et al. (2006) to 0.07%. For Case A the four independent fits give 4.2970×10^{-13} (eq. 6), 4.1923×10^{-13} (eqs. 7–8), 4.1206×10^{-13} (eq. 9) and $4.1699 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ (eq. 11 with T in eV): they agree to within 4% and are all consistent with the $4.2 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ of Gnedin and Madau (2022). Over $10^2\text{--}10^6 \text{ K}$ the largest deviation between eq. (11) and eq. (6) is 3%, which is what identifies eq. (11) as a Case A rate. The ratio $\alpha_B/\alpha_A = 0.603$ at 10^4 K is the fraction of recombinations landing on excited states, and is bracketed by eq. (4) as it must be (Nebrin, 2023). *Not verified:* the underlying Ferland et al. (1992) and Storey & Hummer collisional-radiative calculations themselves, which are taken as given by all the fits above.

Known tensions / open issues, stated explicitly. (i) The definition of R_p used observationally (Eilers et al., 2017) is *not* the Strömgren radius of eq. (1); it is a flux threshold that traces where the quasar radiation dominates the background, which is why R_p is a good clock but a poor

neutral-fraction meter (Eilers et al., 2017; Bolton and Haehnelt, 2007). (ii) The redshift evolution of R_p is measured to be $\propto (1+z)^{-1.44}$ by Eilers et al. (2017) and $\propto (1+z)^{-2.44}$ by Onorato et al. (2025); both are steeper than the $(1+z)^{-1}$ expected at fixed x_{HI} and t_Q (eq. 23), and the two measurements differ from each other, so the redshift scaling should be quoted as observationally uncertain rather than as a prediction. (iii) Bubble radii inferred from Ly α damping wings can exceed the analytic cosmic-average expectation by up to a few dex at a given x_{HI} (Umeda et al., 2023), which is a genuine tension with single-source Strömgren-sphere modelling and points to merged bubbles around clustered sources (Umeda et al., 2023; Neyer et al., 2023; Furlanetto et al., 2004).

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